KAIZEN MATHS

Lesson starter

Limits and First Principles

Behaviour Near a Point

Name _____

SA1

Evaluate each one-sided limit.

$$\lim_{x \to (-2)^-} \frac{|x+2|}{x+2}$$

SA2

Find the one-sided limits and decide whether the two-sided limit exists.

For $f(x) = \{-2x + 4, \quad x4$ find the one-sided limits at $x = 4$ and decide whether $\lim_{x \to 4} f(x)$ exists.

Show your working

KAIZEN MATHS

Lesson starter

Limits and First Principles

Behaviour Near a Point

Name _____

SB1

Use the table values to estimate each limit.

The table shows values of $f(x) = \dfrac{x^2 - 25}{x - 5}$ near $x = 5$. Estimate $\lim_{x \to 5} f(x)$.

x	4.900	4.990	4.999	5.001	5.010	5.100
f(x)	9.900	9.990	9.999	10.001	10.010	10.100

SB2

Evaluate each limit as x tends to infinity.

$$\lim_{x \to \infty} \frac{4x^2 + x + 2}{5x^2 - x - 4}$$

Show your working

KAIZEN MATHS

Exit ticket

Limits and First Principles

Behaviour Near a Point

Name _____

EA1

Evaluate each one-sided limit.

$$\lim_{x \to (-4)^+} \frac{|x+4|}{x+4}$$

Show your working

KAIZEN MATHS

Exit ticket

Limits and First Principles

Behaviour Near a Point

Name _____

EB1

Find the one-sided limits and decide whether the two-sided limit exists.

For $f(x) = \{3x + 5, \quad x4$ find the one-sided limits at $x = 4$ and decide whether $\lim_{x \to 4} f(x)$ exists.

Show your working

KAIZEN MATHS

Exit ticket

Limits and First Principles

Behaviour Near a Point

Name _____

EC1

Use the table values to estimate each limit.

The table shows values of $f(x) = \dfrac{x^2 - 4}{x - 2}$ near $x = 2$. Estimate $\lim_{x \to 2} f(x)$.

x	1.900	1.990	1.999	2.001	2.010	2.100
f(x)	3.900	3.990	3.999	4.001	4.010	4.100

Show your working

KAIZEN MATHS

Exit ticket

Limits and First Principles

Behaviour Near a Point

Name _____

EA1

Evaluate each one-sided limit.

$$\lim_{x \to (-4)^+} \frac{|x+4|}{x+4}$$

Show your working

Teacher answers & working

Limits and First Principles · Behaviour Near a Point

ALG-911D-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

SA1

Evaluate each one-sided limit.

$$\lim_{x\to(-2)^-} \frac{|x+2|}{x+2}$$

Answer:

-1

Working:

This is a one-sided limit, so use the sign of $x+2$ on the left of -2.

As $x \to -2^-$, $x+2 < 0$.

$$\begin{aligned}\frac{|x+2|}{x+2} &= \frac{-(x+2)}{x+2} \\ &= -1\end{aligned}$$

$$\lim_{x\to(-2)^-} \frac{|x+2|}{x+2} = -1$$

SA2

Find the one-sided limits and decide whether the two-sided limit exists.

For $f(x) = \{-2x+4, \quad x4$ find the one-sided limits at $x=4$ and decide whether $\lim_{x\to 4} f(x)$ exists.

Answer:

Does not exist

Working:

Use the rule that applies on the left of 4.

$$\begin{aligned}\lim_{x\to 4^-} f(x) &= \lim_{x\to 4^-} (-2x+4) \\ &= -2(4)+4 \\ &= -4\end{aligned}$$

Use the rule that applies on the right of 4.

$$\begin{aligned}\lim_{x\to 4^+} f(x) &= \lim_{x\to 4^+} (-x+3) \\ &= -(4)+3 \\ &= -1\end{aligned}$$

The left-hand and right-hand limits are different, so the two-sided limit does not exist.

$\lim_{x\to 4} f(x)$ **does not exist**

Teacher answers & working

Limits and First Principles · Behaviour Near a Point

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SB1

Use the table values to estimate each limit.

The table shows values of $f(x) = \dfrac{x^2 - 25}{x - 5}$ near $x = 5$. Estimate $\lim_{x \to 5} f(x)$.

x	4.900	4.990	4.999	5.001	5.010	5.100
f(x)	9.900	9.990	9.999	10.001	10.010	10.100

Answer:

10

Working:

The values of x approach 5 from both sides.

The corresponding values of $f(x)$ approach 10.

Algebra confirms the table pattern.

$$\lim_{x \to 5} \frac{x^2 - 25}{x - 5} = \lim_{x \to 5} \frac{(x - 5)(x + 5)}{x - 5}$$
$$= \lim_{x \to 5} (x + 5)$$
$$= 10$$

SB2

Evaluate each limit as x tends to infinity.

$$\lim_{x \to \infty} \frac{4x^2 + x + 2}{5x^2 - x - 4}$$

Answer:

$\dfrac{4}{5}$

Working:

The highest power in the numerator and denominator is x^2, so divide every term by x^2.

$$\lim_{x \to \infty} \frac{4x^2 + x + 2}{5x^2 - x - 4} = \lim_{x \to \infty} \frac{4 + \frac{1}{x} + \frac{2}{x^2}}{5 + \frac{-1}{x} + \frac{-4}{x^2}}$$

As $x \to \infty$, the terms $\dfrac{1}{x}$ and $\dfrac{1}{x^2}$ tend to zero.

$$\lim = \frac{4 + 0 + 0}{5 + 0 + 0}$$
$$= \frac{4}{5}$$

Teacher answers & working

Limits and First Principles · Behaviour Near a Point

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EA1

Evaluate each one-sided limit.

$$\lim_{x\to(-4)^+} \frac{|x+4|}{x+4}$$

Answer:

1

Working:

This is a one-sided limit, so use the sign of $x+4$ on the right of -4.

As $x \to (-4)^+$, $x+4>0$.

$$\begin{aligned}\frac{|x+4|}{x+4} &= \frac{x+4}{x+4}\\ &= 1\end{aligned}$$

$$\lim_{x\to(-4)^+} \frac{|x+4|}{x+4} = 1$$

EB1

Find the one-sided limits and decide whether the two-sided limit exists.

For $f(x) = \{3x+5, \quad x4$ find the one-sided limits at $x=4$ and decide whether $\lim_{x\to4} f(x)$ exists.

Answer:

Does not exist

Working:

Use the rule that applies on the left of 4.

$$\begin{aligned}\lim_{x\to4^-} f(x) &= \lim_{x\to4^-}(3x+5)\\ &= 3(4)+5\\ &= 17\end{aligned}$$

Use the rule that applies on the right of 4.

$$\begin{aligned}\lim_{x\to4^+} f(x) &= \lim_{x\to4^+}(x+18)\\ &= (4)+18\\ &= 22\end{aligned}$$

The left-hand and right-hand limits are different, so the two-sided limit does not exist.

$\lim_{x\to4} f(x)$ **does not exist**

Teacher answers & working

Limits and First Principles · Behaviour Near a Point

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EC1

Use the table values to estimate each limit.

The table shows values of $f(x) = \dfrac{x^2 - 4}{x - 2}$ near $x = 2$. Estimate $\lim\limits_{x \to 2} f(x)$.

x	1.900	1.990	1.999	2.001	2.010	2.100
f(x)	3.900	3.990	3.999	4.001	4.010	4.100

Answer:

4

Working:

The values of x approach 2 from both sides.

The corresponding values of $f(x)$ approach 4.

Algebra confirms the table pattern.

$$\begin{aligned}\lim_{x \to 2} \frac{x^2 - 4}{x - 2} &= \lim_{x \to 2} \frac{(x - 2)(x + 2)}{x - 2} \\ &= \lim_{x \to 2} (x + 2) \\ &= 4\end{aligned}$$